

VOLUME OF RIGHT RECTANGULAR PRISMS



LESSON INTRODUCTION

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational-number edge lengths using a visual model and a formula.

LEARNING TARGETS

- I can pack a rectangular prism using a cube with fractional edges.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.3.3 Solve real-world problems involving the volume of right rectangular prisms, including problems with an unknown edge length, with whole-number edge lengths using a visual model or a formula. Write an equation with a variable for the unknown to represent the problem.

WORKING TOWARD

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

ESSENTIAL QUESTIONS

- Why is volume measured in cubic units?
- What is a unit cube?
- How can you use a unit cube to find the volume of a rectangular prism?
- What is the formula to find the volume of a rectangular prism?

LESSON NARRATIVE

In this lesson, students investigate finding the volume of a right rectangular prism with fractional edge lengths by packing prisms with unit cubes of the appropriate unit fraction edge lengths. Students discover that finding the volume using this method produces the same result as finding the volume with the formula. The lesson begins with an exploration on packing rectangular prisms with cubes, but the bulk of the lesson focuses on guided instruction and practice in finding volume of right rectangular prisms and writing appropriate units.

COMPONENT SUMMARY

8.6.1 WARM-UP (~5 MIN.)

Representations and descriptions of volume

8.6.2 EXPLORATION (5 MIN.)

Explore volume and dimensions of the unit cube

8.6.3 EXPLORATION (10-15 MIN.)

Explore volume of cubes with fractional side lengths by comparing to the unit cube

8.6.4 GUIDED INSTRUCTION (20-25 MIN.)

Students calculate the volume of rectangular prisms with unit cubes and compare results with the volume formula.

8.6.5 YOUR TURN! (5-10 MIN.)

Independent practice finding the volume of a rectangular prism with fractional side lengths

8.6.6 WRAP-UP (~5 MIN.)

Students match prisms with correct dimensions and volume

RESOURCES

GLOSSARY

Key Terms:

- Cube

Additional Terms:

- Volume
- Length, Width, Height
- Unit cube

MATERIALS

- (Optional) Unit cube manipulatives
- (Optional) Paper
- (Optional) Rulers
- (Optional) Scissors

ADDITIONAL PREPARATION

- There is a Desmos activity builder to accompany this lesson. If using the Desmos version of the activity, review slides ahead of time.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students with gaps in spatial awareness may struggle with composing and decomposing rectangular prisms into fractional cubes.
- Students may struggle to differentiate between two-dimensional faces and three-dimensional cubes.

THINGS TO CONSIDER

- Facilitate class discussion on 8.6.3 Exploration question 3, which could prove helpful for students as they visualize decomposing a larger rectangular prism into smaller cubes.
- Consider having physical unit cubes available for students to manipulate into larger prisms.
- The conceptual foundation that the lesson lays should not be rushed. Allow students to explore and ask questions. Once students understand conceptually how to break a cube into fractional edges, then the mathematics makes sense. Utilizing the Desmos activity will help students visualize the concepts. Students may go through the Desmos activity faster than the workbook lesson. Determine if students should record their answers from the Desmos activity into the workbook.
- If time is an issue, assign 8.6.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

8.6.3 Exploration intentionally sets the stage for students to think about the relationship between unit cubes and the volume of a larger prism. This low-stakes task makes the exploration accessible to all students and provides opportunities for students to share what they notice and wonder about volume and unit cubes.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

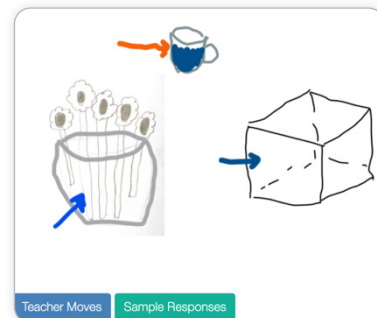
MA.K12.MTR.2.1 Multiple Representations

In this lesson, students are actively engaged building multiple representations to demonstrate understanding in more than one way, including through modeling and manipulatives. Utilize 8.6.3 Exploration and 8.6.4 Guided Instruction questions to position students to recognize volume with a visual representation and the formula by connecting the process of packing prisms with cubes and the volume formula.

DIFFERENTIATE INSTRUCTION

This lesson focuses heavily on spatial reasoning and promotes visual representations of volume of rectangular prisms using unit cubes. Providing physical manipulatives or tools to draw visual models in several lesson activities (8.6.3 Exploration, 8.6.4 Guided Instruction, and 8.6.5 Your Turn!) can allow additional entry points for kinesthetic and visual learners.

Additionally, students can engage with each other synchronously or asynchronously using the accompanying Desmos Activity Builder for this lesson. Consider leveraging the digital version of this lesson before, during, or after class for additional entry points to the learning.



8.6.1 WARM-UP: GUESS THE WORD

Component Overview: Students observe three visual representations of volume and choose statements that best describe volume. Consider using Notice and Wonder routines as a group and write students' responses in a shared classroom space. Look for general themes from their responses, like volume, area, 3D object, and/or content in an object. Prompt responses within these themes and ask students to briefly elaborate.

ENRICHMENT

As students look at the math representations that Micah drew to represent volume (which show the space inside a cup, flowerpot, and cube), encourage mathematical reasoning about volume by asking questions like:

- What do you notice about the drawings? What do you wonder?
- What do you think each drawing represents? What is the same or different about these objects in the real-world?
- Which drawings represent volume? How can you tell?

QUESTIONING

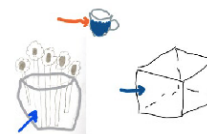
Early finishers can create their own drawing to represent volume, then compare their drawing to Micah's or another student's work. This activity can also be extended by having students identify what mathematical concept is described by the three incorrect answers in question 1.



8.6.1 Warm-Up: Guess the Word

Micah and his family were playing a game in which each person chooses a card with a word written on it. Once a person chooses, they have one minute to draw a picture of the word. Then, his team guesses the word.

Micah chose the word "volume". His drawing is shown.



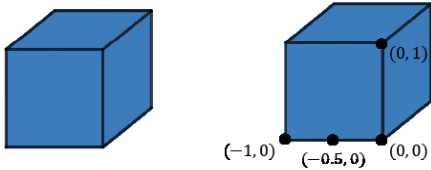
1. Choose the statement(s) that best describe volume. Select all that apply.

- ☐ the sum of the length of all the edges of a three-dimensional object
- ☐ the distance around a two-dimensional shape
- ☒ the amount of three-dimensional space enclosed within a container
- ☐ the total area of the surface of a three-dimensional object
- ☐ the formula length times width times height



8.6.2 Exploration: The Unit Cube

1. Two images of the same cube are shown. The image on the right shows the coordinates of some of the edges of the cube.



Share three things you know, two things you are not sure of, and one thing you have a question about.

I Know...	I'm Not Sure Of...	A Question I Have...
Answers will vary. A cube is a 3D shape. A cube is made of squares. A cube has six sides.	Answers will vary. How to find the length of a side. Which points to use to find the side lengths.	Answers will vary. How can I find out how much space is inside the cube?

2. This cube is known as a unit cube because each dimension (length, width, and height) is 1 unit.

What is the volume of the cube? Include the units for the cube's volume.

1 unit³



A cube is a rectangular prism with six congruent square faces.

8.6.2 EXPLORATION: THE UNIT CUBE

Component Overview: Students make observations about the unit cube's dimensions and explore its volume. Consider having partners complete questions 1 and 2 for about five minutes while writing students' observations and questions on a shared classroom space, then regroup and revisit questions from question 1 at the end of 8.6.2 Exploration.

TEACHER MOVES

For question 1, consider highlighting the most common responses for each category, in particular the questions students have. Consider posing some questions or aspects students are unsure about to the whole group to gather students' initial thoughts, then prompt students to attempt answering the questions at the end of the 8.6.2 Exploration.

Emphasize what is considered a unit cube and why it is a unit cube. This helps students then relate to fractional cubes. If manipulatives are available, then consider showing one cube as the unit cube.

ENRICHMENT

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8.6.3 EXPLORATION: DRAWING FRACTIONAL CUBES

TEACHER MOVES

Notice and Wonder

Use these first two questions as opportunities to have students discuss the relationship between side lengths and volume. Some students may recall the volume formula (length $\times$ width $\times$ height).

Component Overview: Students explore how to decompose the unit cube into cubes with fractional side lengths of $\frac{1}{2}$ and discover different ways to visualize the total number of these cubes contained within a unit cube. Consider using Notice and Wonder routines as a class with questions 1-3 to highlight important concepts related to volume, then have partners continue working together to complete question 4.

ENRICHMENT

- How many half unit cubes did you guess would fit in a unit cube? How did you make your guess?
- How much larger do you think the volume of the unit cube is compared to the half unit cube?

ENRICHMENT

How do you know there are four cubes in each level?

DIFFERENTIATE INSTRUCTION

Consider allowing students who struggle with spatial reasoning to use cube manipulatives to physically pack a cube or build a side-by-side view of a unit cube and a fraction cube. If physical cubes are unavailable, students can also use the Desmos activity to virtually pack the cube.

QUESTIONING

Students who finish early could extend their thinking by answering the following prompts:

- Would any other cube size fit inside the unit cube?
- Choose a different dimension cube (e.g., $\frac{1}{3}$ unit) and find how many of those smaller cubes can be packed into the unit cube.
- Can you come up with a pattern or rule to find the total number of cubes in a unit cube? What is the pattern or rule? How do you know it always works?



8.6.3 Exploration: Drawing Fractional Cubes

1. Two cubes are shown.

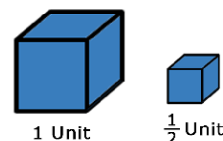
The cube on the left is a unit cube. The cube on the right has different dimensions.



Share an observation and something you wonder about this new cube.

Observation	Wondering
Answers will vary. The cube on the right is smaller than the cube on the left. Both cubes have the same amount of sides.	Answers will vary. How many small cubes can fit in the big cube?

2. A unit cube and a half-unit cube are shown.



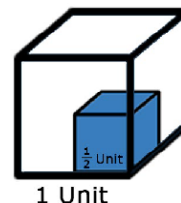
Make a guess of how many cubes with a length of $\frac{1}{2}$ unit, width of $\frac{1}{2}$ unit, and height of $\frac{1}{2}$ unit will fit in the 1-unit cube.

Answers will vary. 2, 4, 8

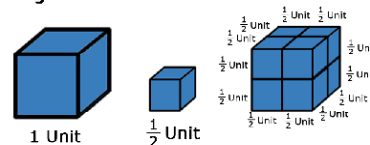
3. The figure shows a half-unit cube inside a unit cube.

Explain how the figure helps you to visualize how to pack cubes that have a fractional length.

Answers will vary. It looks like I make two layers of 4 for a total of 8 half-unit cubes.



4. Three figures are shown.



- a. Explain how the figures help you visualize how to pack cubes that have a fractional length.

Answers will vary.

I was able to see how the cubes were half of the side lengths. It reminded me of a 2x2 Rubik's Cube.

- b. To find the number of cubes whose dimensions are $\frac{1}{2}$ unit that fit into a cube whose dimensions are 1 unit, Derek wrote 1 as $\frac{2}{2}$ so he could see that $\frac{1}{2} + \frac{1}{2}$ would add to length of $\frac{2}{2}$ or 1. Complete his explanation.

Since $\frac{1}{2} + \frac{1}{2} = \frac{2}{2}$, or 1, I need 2 cubes for the length, 2 cubes for the width, and 2 cubes for the height.

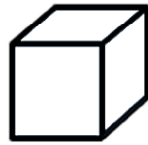
Since 2 $\times$ 2 $\times$ 2 = 8 I know that I need 8 cubes with dimensions $\frac{1}{2}$ unit to fit into one 1-unit cube.



8.6.4 Guided Instruction: Volume with Fractional Cubes

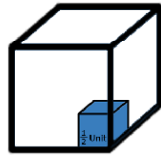
The cube shown is a scale model of a moving box.

1. The edges of the box are $1\frac{1}{2}$ feet long. How many cubes, whose dimensions are $\frac{1}{2}$ feet, can fit inside each dimension of the moving box?
3



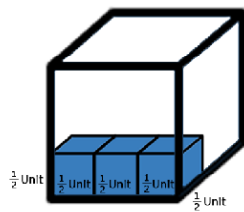
$1\frac{1}{2}$ Units

2. Sketch the cubes that can fit inside each dimension of the moving box.
Check student drawings for understanding. Each side of the cube should be divided into 9 squares.



$1\frac{1}{2}$ Units

3. Since 3 cubes, whose dimensions are $\frac{1}{2}$ unit, will fit inside each dimension, how many of these cubes would be needed to pack the moving box?
27



$1\frac{1}{2}$ Units

8.6.4 GUIDED INSTRUCTION: VOLUME WITH FRACTIONAL CUBES

Component Overview: This section guides students through the process of decomposing prisms into cubes to calculate and recognize a representation of volume. Students confirm this method by applying the volume formula for a rectangular prism ($V = lwh$). Consider guiding students through each question in this section as a class, pausing to discuss questions after each question.

TEACHER MOVES

For this activity, consider demonstrating directly how to sketch the cubes inside the larger prism.

Have students work in partners to check one another's sketches. It is important that students do not overcomplicate the sketching process. Instead, sketching helps them to quickly see how many cubes are in a layer. Some students may find a ruler is helpful while sketching.

ENRICHMENT

Prompt students to consider how the cubes relate to volume by asking:

- How many squares can you see on each side of the cube?
- Is this the same as the total number of cubes packed inside? Why or why not?

TEACHER MOVES

Consider asking students to discuss multiple pathways to discover that 27 cubes are needed to pack the box. Allow extra time for this discussion because it is important for students to understand how to arrive at a total number of cubes.

8.6.4 GUIDED INSTRUCTION: VOLUME WITH FRACTIONAL CUBES (CONTINUED)

ENRICHMENT

To scaffold thinking for question 4, encourage students to revisit 8.6.3 Exploration, prompting them to consider:

- What would the volume be for a cube whose dimensions are 1 foot? Why do we call it a unit cube?
- How many half cubes were able to fit inside one unit cube?
- What is the volume of a single half cube? How do you know?

TEACHER MOVES

Ensure that students understand that any of the three dimensions could be labeled as length. The “why” is easily visualized by rotating a cube or rectangular prism so that the side that is on the bottom varies. Practice using various labels to help students relate to this flexibility.

ENRICHMENT

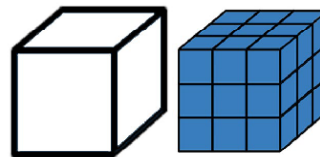
Strengthen student understanding of how to use the formula to find volume in questions 6 and 7 by asking:

- What do l, w, and h represent in the formula?
- What is the length of the cube? Width? Height?
- How do we substitute values into this formula?

DIFFERENTIATE INSTRUCTION

In question 8, position students to note the length, width, and height of the larger prism (in addition to the total number of cubes). This sets students up to be able to use these dimensions in calculating volume. Have students label the dimensions on the prisms.

4. This cube is a scale model of a moving box. The edges of the box are $1\frac{1}{2}$ feet long.



Since there are 27 cubes whose dimensions are $\frac{1}{2}$ feet that will fill the moving box, the volume of the moving box can be found by multiplying the volume of the fractional cube by 27. What is the volume of a cube whose dimensions are $\frac{1}{2}$ feet?

$$\frac{1}{8} \text{ ft}^3$$

5. Since there are 27 cubes whose volume is $\frac{1}{8}$ cubic feet that will fill in the moving box, the volume of the moving box can be found by multiplying 27 and $\frac{1}{8}$ cubic feet. What is the volume of a cube of the moving box?

$$3\frac{3}{8} \text{ ft}^3$$

6. Find the volume of the moving box using the formula for volume of a rectangular prism or cube.

$$V = lwh$$

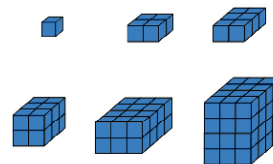
Each dimension of the box is $1\frac{1}{2}$ feet.

$$V = lwh = \frac{3}{2} \times \frac{3}{2} \times \frac{3}{2} = \frac{27}{8} = 3\frac{3}{8} \text{ ft}^3$$

7. Does the volume you found using the formula match the volume you found by counting the number of fractional cubes?

Yes

8. The images show different rectangular prisms composed of various numbers of cubes.



- a. What do you notice?

Answers will vary. If I multiply the amount of cubes that make up each side length, it equals how many cubes are shown total.

- b. Explain how the dimensions of the prism can be determined using the cubes shown if each small blue cube is 1 unit cube.

You can count how many cubes make up each side to use for length, width, and height to find the volume.

9. The diagram shows 3 cubes whose dimensions are $\frac{1}{4}$ inches.



Select the statements that correctly describe the volume of the diagram as shown.

- ☐ Each cube has a volume of $\frac{1}{4}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{16}$ cubic inches.
- ☒ Each cube has a volume of $\frac{1}{64}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{4}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{16}$ cubic inches.
- ☒ The volume of all three cubes is $\frac{3}{64}$ cubic inches.

10. The same diagram of 3 cubes whose dimensions are $\frac{1}{4}$ inches is shown.



- a. The dotted line represents the total length of the 3 cubes. How long, in inches, is the dotted line?

- A $\frac{4}{4}$ inches
- B $\frac{1}{4}$ inches
- ☒ C $\frac{3}{4}$ inches
- D $\frac{2}{4}$ inches

- b. The blue line represents the width of the 3 cubes.

What is the width, in inches?

- ☒ A $\frac{1}{4}$ inches
- B $\frac{2}{4}$ inches
- C $\frac{3}{4}$ inches
- D $\frac{4}{4}$ inches

- c. The green line represents the height of the cubes.

What is the height, in inches?

- ☒ A $\frac{1}{4}$ inches
- B $\frac{2}{4}$ inches
- C $\frac{3}{4}$ inches
- D $\frac{4}{4}$ inches

- d. Use the dimensions you found to determine the volume of all three cubes with the formula, $V = lwh$.

$$V = 3 \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{3}{64} \text{ in}^3$$

Finding volume using fractional unit cubes can be done by first finding the volume of one of the fractional unit cubes, and then multiplying that by the total number of fractional unit cubes.

8.6.4 GUIDED INSTRUCTION: VOLUME WITH FRACTIONAL CUBES (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider allowing students to work in partner pairs for questions 9-11. Circulate the room and monitor for correct application of the volume formula. Student approaches might illuminate potential gaps (e.g., multiplying fractional side lengths to find volume).

As a visual entry point to question 9, consider prompting students to draw arrows to the diagram or to arithmetic work as evidence on why they know the statements are correct.

ENRICHMENT

For question 10, deepen conceptual understanding by eliciting responses to the following:

- Explain why the volume of each cube is $\frac{1}{64} \text{ in}^3$.
- How can you find the volume of the entire prism once you know the volume of one cube?

DIFFERENTIATE INSTRUCTION

For questions 10 and 11 and into the practice in the next section (8.6.5 Your Turn!), consider providing a template or graphic for students who may struggle with organization. The organizer can have prompts to substitute into the volume formula, reminders on how to multiply fractional values, and tips on converting from improper to mixed-number form.

8.6.4 GUIDED INSTRUCTION: VOLUME WITH FRACTIONAL CUBES (CONTINUED)

ENRICHMENT

For question 11a, if a student multiplied 24 by $\frac{1}{4}$ to find the volume of the prism, would the student arrive at the correct volume? Why or why not?

TEACHER MOVES

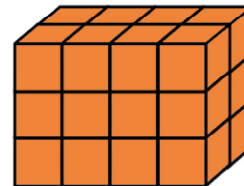
Encourage synthesis of the learning targets in this section by positioning students to recall the two strategies to finding volume. Consider using Turn and Talk routines to have students summarize the two strategies to finding the volume of rectangular prisms with fractional side edges.

- Strategy 1: Choose a unit cube that fits into each dimension. Multiply the total number of cubes that pack the prism by the volume of the unit cube.
- Strategy 2: Apply the volume formula, $V = lwh$.

Additionally, consider having students share which strategy they prefer and why. Use this reflection to highlight strategic approaches to finding volume, including when it may be better to use the formula versus when it may be easier to use unit cubes.

11. A rectangular prism is shown.

- a. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?
24



- b. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches.

Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

$$\frac{3}{8} \text{ in}^3$$

- c. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$. Show your work and your answer.

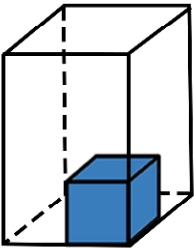
$$V = (3)\frac{1}{4}(4)\frac{1}{4}(2)\frac{1}{4}$$

$$V = \frac{3}{8} \text{ in}^3$$



8.6.5 Your Turn!

1. A rectangular prism with one cube inside is shown.



- a. How many cubes will it take to fill the rectangular prism?
12

- b. The rectangular prism has a length of $\frac{16}{3}$ centimeters (cm), width of $\frac{16}{3}$ cm, and height of 8 cm as shown. Use the number of cubes in each dimension to determine the dimensions of the fractional unit cube.

height = $8 \div 3$, length and width = $\frac{16}{3} \div 2$

- c. Complete the statement.

The unit cube shown has dimensions of $\frac{8}{3}$ cm. It will take **12** of these unit cubes to fill the rectangular prism.

- d. Explain how to find the volume of the rectangular prism using the information from Part C.
To find the volume of the rectangular prism using the information from Part C, find the volume of the unit cube by multiplying the dimensions 3 times. Then multiply the volume of the unit cube by the number of cubes it will take to fill the rectangular prism.

$\frac{8}{3} \times \frac{8}{3} \times \frac{8}{3} \times 12 = 227\frac{5}{9} \text{ cm}^3$

8.6.5 YOUR TURN!

Component Overview: Students work independently to calculate the volume of a rectangular prism with fractional edges by packing the prism with cubes. While some of the class completes this work independently, consider pulling a small group of students who may need additional support in using the strategies to find volume from the previous section.

DIFFERENTIATE INSTRUCTION

Students may still struggle with differentiating between the volume of a single unit cube and the volume of the entire prism. Consider using this section as an opportunity to review strategies to find the total number of cubes and strategies to determine the dimensions of the cube with a small group of students who need additional support.

QUESTIONING

Students who finish early can confirm their answer in question 1d by applying the volume formula and comparing solutions.



8.6.6 Wrap-Up: Ticket Out the Door

1. Match each image with its corresponding dimensions and volume. Use the table below to record your answers.

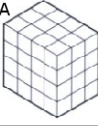
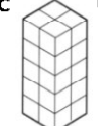
Image	Dimensions	Volume
A 	The edges of each cube are $3\frac{1}{4}$ in. long. D1 Length - 5 in. Width - 5 in. Height - 12.5 in.	V1 Volume $5\frac{1}{3}$ cubic in.
B 	The edges of each cube are $\frac{2}{3}$ in. long. D2 Length - 13 in. Width - 9.75 in. Height - 13 in.	V2 Volume 1,647.75 cubic in.
C 	The edges of each cube are 2.5 in. long. D3 Length - 2 in. Width - $(\frac{2}{3})$ in. Height - 2 in.	V3 Volume 312.5 cubic in.

Image	Dimensions	Volume
A	D2	V2
B	D3	V1
C	D1	V3

8.6.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: Students complete a match activity that assesses their ability to identify the dimensions of a rectangular prism with fractional side lengths and calculate volume. This Wrap-Up is composed of three multistep problems that require proficiency in multiplying fractional values, determining dimensions of a rectangular prism, and calculating the volume of a rectangular prism.

DIFFERENTIATE INSTRUCTION

For students who typically require additional time to complete calculations, consider removing image C from the required task so students can focus on correctly calculating the volume for images A and B.